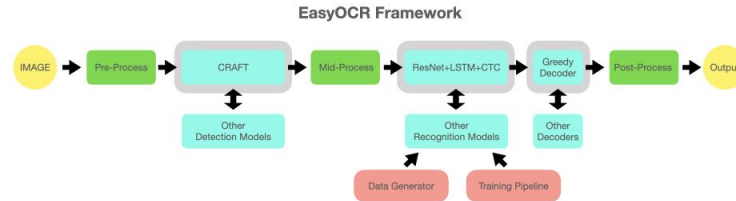
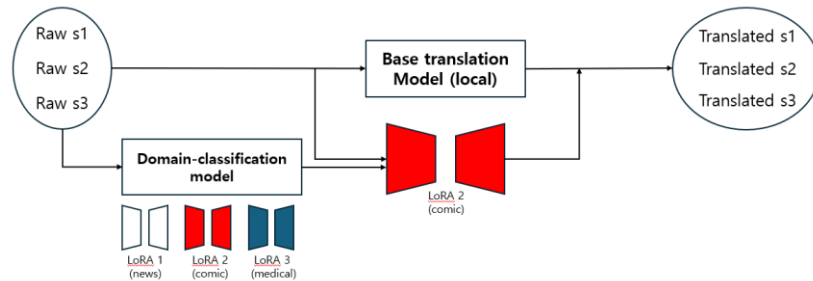
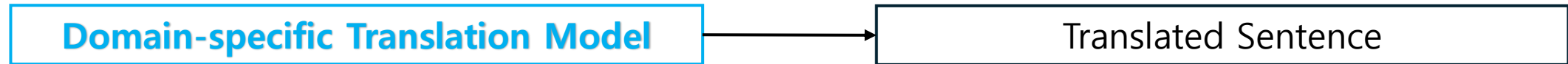


Local Image Translation Program

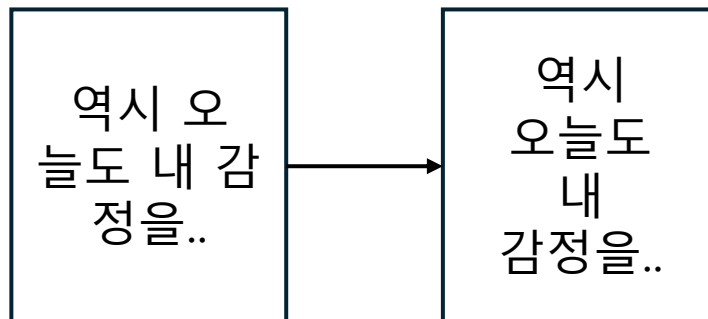
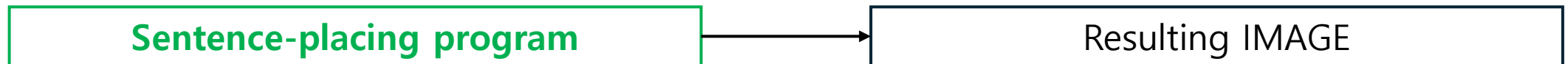
Sinyoung Kwon, Seoul National University



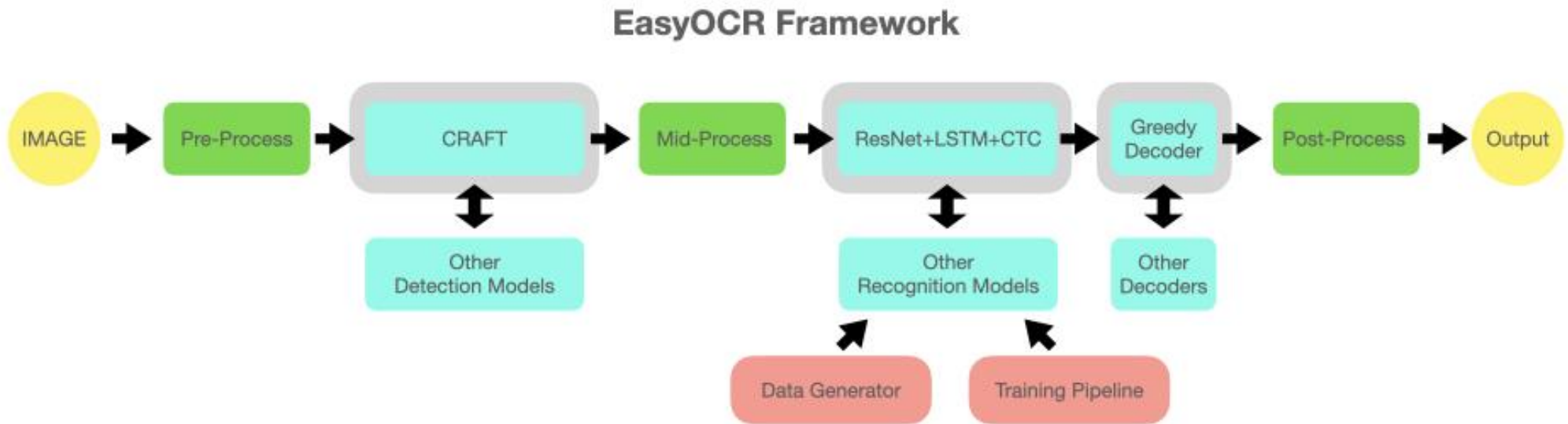
やはり今日も俺の感情を..
二度と俺に...



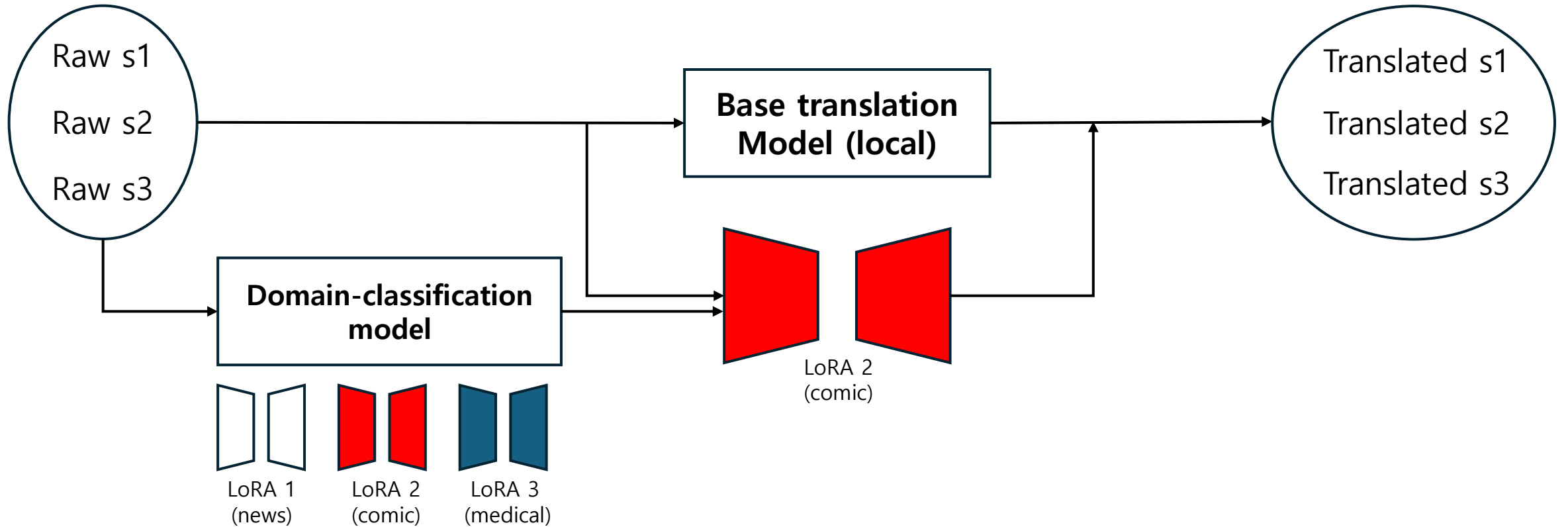
역시 오늘도 내 감정을..
두 번 다시 내게..



EasyOCR Framework



Domain-specific Translation Model



Select Base Translation model

- Choose a small model that can be run locally
- (Optional) fine-tuning with large dataset

Ex) OPUS-MT, Helsinki-NLP..

Train LoRA

- LLMs already have great performance; so if we prompt well for the domain, it produces great translation results at a level that does not require human post-processing.
- Using **transfer learning**, train LoRA domain-specifically.
- In actual translation phase, use **domain-classification model** to determine which LoRA to use.
 - I don't think it's going to be that hard.
 - Comics, medical, ... (default LoRA)

Benchmarking Service: translatemanga.net

- <https://translatemanga.net/en>
- Drawbacks
 1. The translation is not completely smooth (little improvement seems possible)
 2. The translation sentence placing, font size, etc. (not neat, requiring human intervention)
 3. LLM API call base, external service dependent
- Differences from my project
 1. (Goal) more natural translation
 2. Appropriate font selection and placement location considering spacing of translated sentences, etc
 3. Run on Local